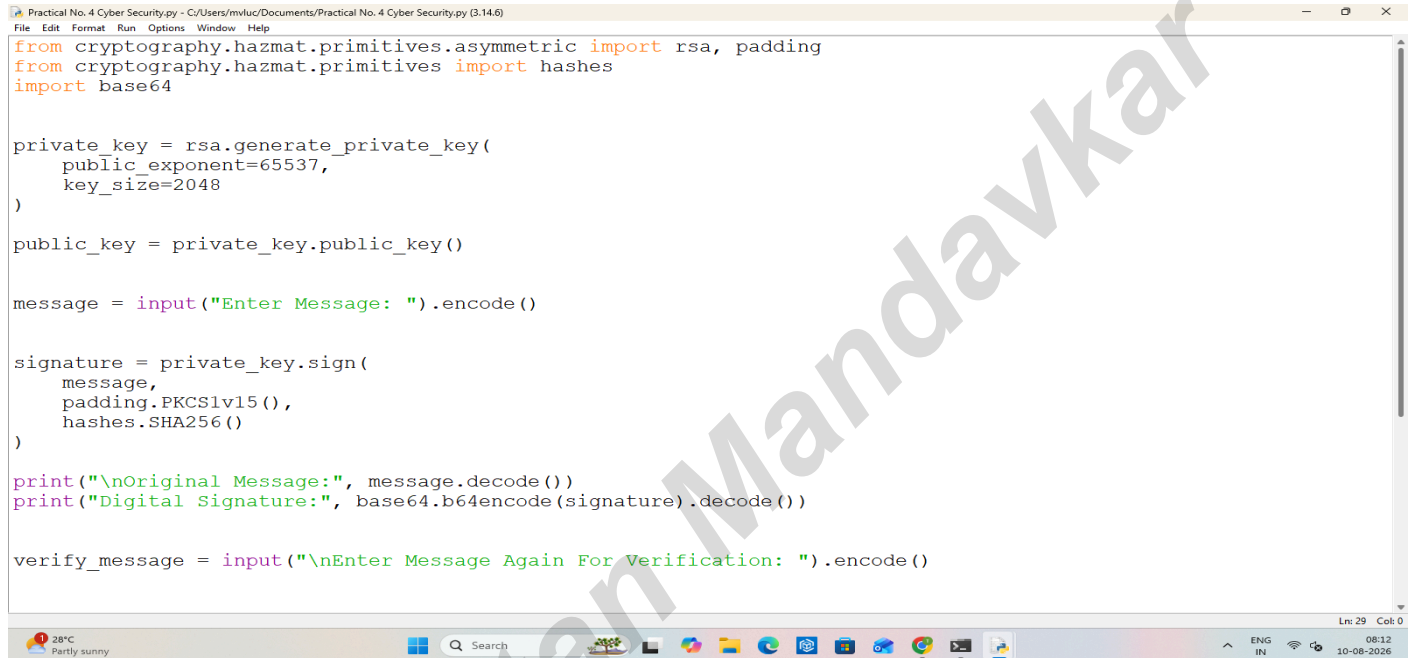


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Aim:- Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

CLI MODE:-



```
from cryptography.hazmat.primitives.asymmetric import rsa, padding
from cryptography.hazmat.primitives import hashes
import base64

private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)

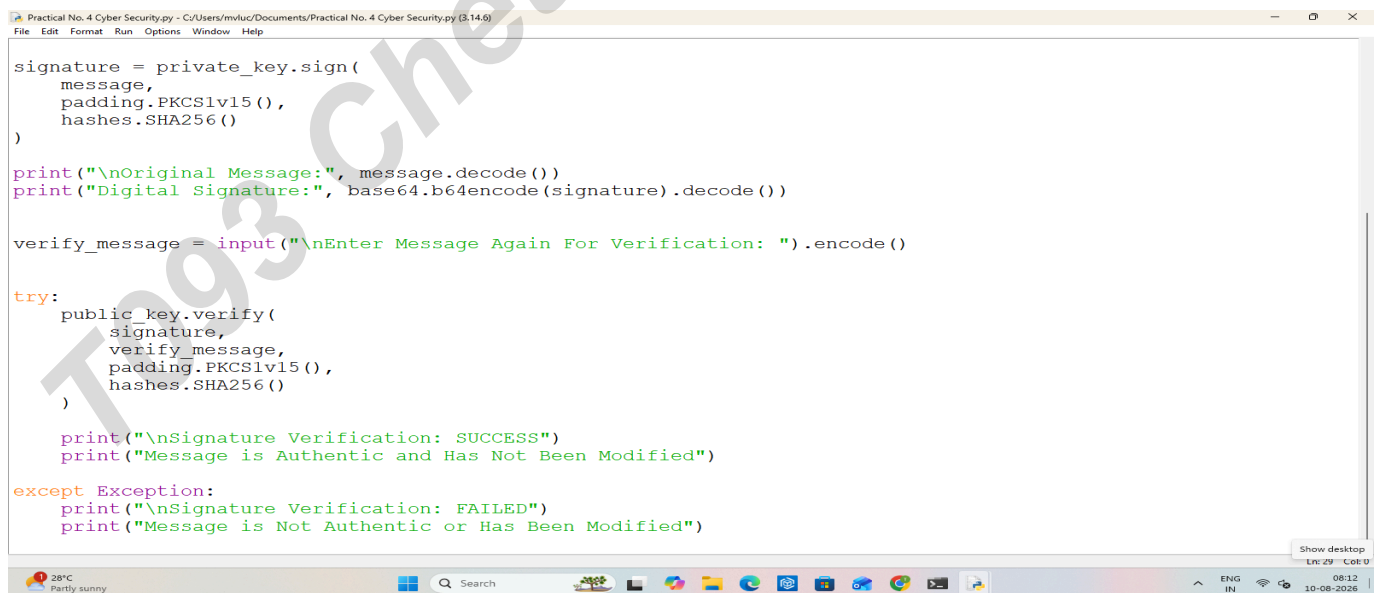
public_key = private_key.public_key()

message = input("Enter Message: ").encode()

signature = private_key.sign(
    message,
    padding.PKCS1v15(),
    hashes.SHA256()
)

print("\nOriginal Message:", message.decode())
print("Digital Signature:", base64.b64encode(signature).decode())

verify_message = input("\nEnter Message Again For Verification: ").encode()
```



```
signature = private_key.sign(
    message,
    padding.PKCS1v15(),
    hashes.SHA256()
)

print("\nOriginal Message:", message.decode())
print("Digital Signature:", base64.b64encode(signature).decode())

verify_message = input("\nEnter Message Again For Verification: ").encode()

try:
    public_key.verify(
        signature,
        verify_message,
        padding.PKCS1v15(),
        hashes.SHA256()
    )

    print("\nSignature Verification: SUCCESS")
    print("Message is Authentic and Has Not Been Modified")
except Exception:
    print("\nSignature Verification: FAILED")
    print("Message is Not Authentic or Has Been Modified")
```

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Output:-



```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py =====
Enter Message: Welcome To Digital Signature Practical

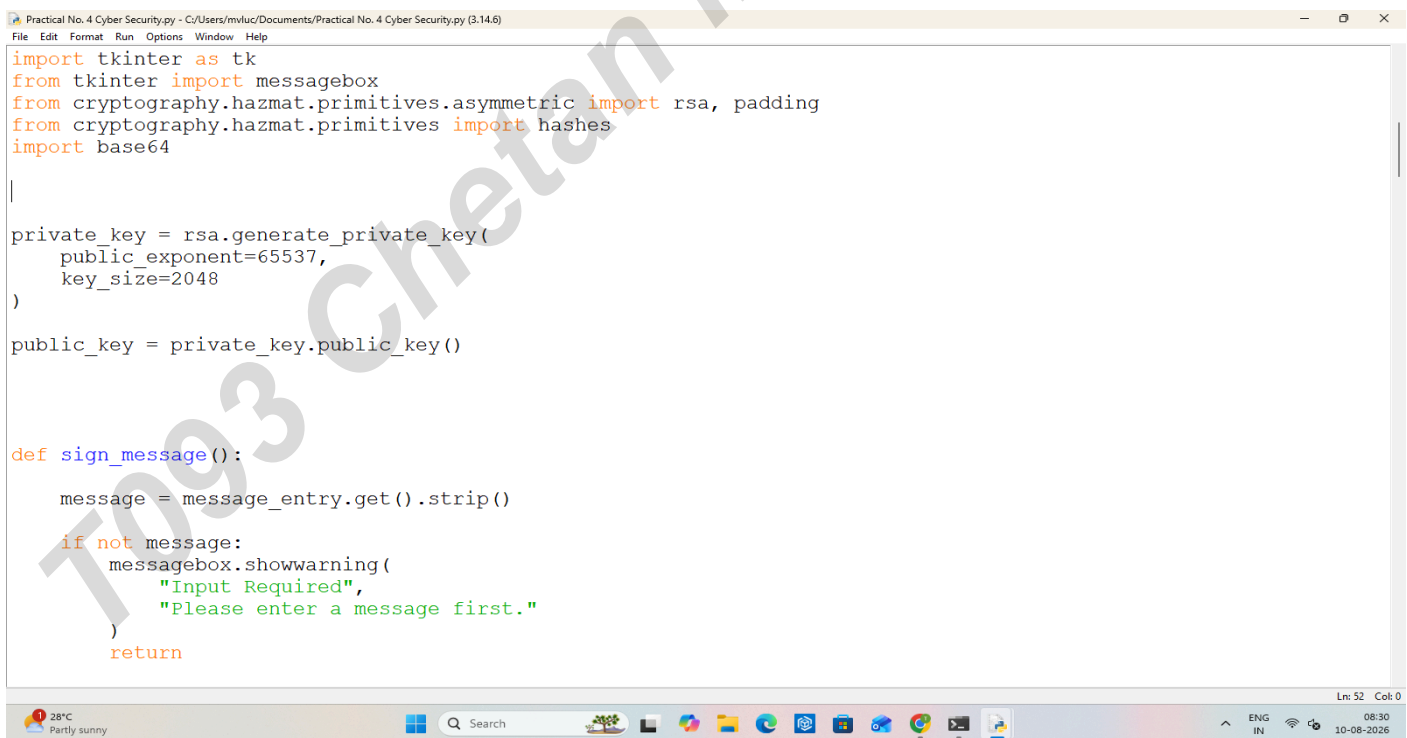
Original Message: Welcome To Digital Signature Practical
Digital Signature: pUHKKEBfGog0LfjsMJdNkRrYFGwEMsm6jpySyQIzBipCYu4vL7Wq5JMBYDRpWAqe0KQGm3gAlGDYU+6E30k2s6AbL
GXJVF43AQjSjbrUAG4jXMSAQlIDzvxtm5Gmm7m+Fm4vt2f4ZdBizj0yGsysMz7ErxzgASZJExZMAM/RDFZ+zP5BXpwlq5FZoy7WScXHL2T9o
8Ani2e5zoJBLhbWQ5bnFekf97axrAhpdxlQ5PEuFSRFIPB+Mhmv+eWMyjrlIeCMUt6le4SXeFV3xdVt+/1L5I9REn83PYystvuiEczlzGkp
HZp77BleFWKMX5Y40P388wJwG/dE/Gs/WbC+g==

Enter Message Again For Verification: Digital Practical

Signature Verification: FAILED
Message is Not Authentic or Has Been Modified

>>>
```

GUI VERSION:-



```
Practical No. 4 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

import tkinter as tk
from tkinter import messagebox
from cryptography.hazmat.primitives.asymmetric import rsa, padding
from cryptography.hazmat.primitives import hashes
import base64

private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)

public_key = private_key.public_key()

def sign_message():
    message = message_entry.get().strip()

    if not message:
        messagebox.showwarning(
            "Input Required",
            "Please enter a message first."
        )
    return
```

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```
Practical No. 4 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

message_bytes = message.encode()

signature = private_key.sign(
    message_bytes,
    padding.PKCS1v15(),
    hashes.SHA256()
)

signature_text = base64.b64encode(signature).decode()

signature_entry.delete("1.0", tk.END)
signature_entry.insert(tk.END, signature_text)

status_label.config(
    text="● SIGNATURE GENERATED SUCCESSFULLY",
    fg="#00e676"
)

status_detail.config(
    text="The message has been digitally signed using the RSA private key.",
    fg="#9aa4b2"
)

def verify_signature():

    message = message_entry.get().strip()

    signature_text = signature_entry.get("1.0", tk.END).strip()
```

```
Practical No. 4 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

signature_text = signature_entry.get("1.0", tk.END).strip()

if not message:
    messagebox.showwarning(
        "Input Required",
        "Please enter a message."
    )
    return

if not signature_text:
    messagebox.showwarning(
        "Signature Required",
        "Please generate or enter a digital signature."
    )
    return

try:
    signature = base64.b64decode(signature_text)
    public_key.verify(
        signature,
        message.encode(),
        padding.PKCS1v15(),
        hashes.SHA256()
    )

    status_label.config(
```

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```
Practical No. 4 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

    status_label.config(
        text="✓ SIGNATURE VERIFIED",
        fg="#00e676"
    )

    status_detail.config(
        text="Message is authentic and has not been modified.",
        fg="#9aa4b2"
    )

except Exception:

    status_label.config(
        text="✗ SIGNATURE VERIFICATION FAILED",
        fg="#ff5252"
    )

    status_detail.config(
        text="Message is not authentic or has been modified.",
        fg="#9aa4b2"
    )

def clear_fields():

    message_entry.delete(0, tk.END)

    signature_entry.delete("1.0", tk.END)
```

Ln: 52 Col: 0
28°C Partly sunny 08:31 10-08-2026

```
Practical No. 4 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

    status_label.config(
        text="● READY",
        fg="#00bcd4"
    )

    status_detail.config(
        text="Enter a message and generate a digital signature.",
        fg="#9aa4b2"
    )

root = tk.Tk()

root.title("RSA Digital Signature")
root.geometry("780x650")

root.configure(bg="#0b1220")
root.resizable(False, False)

header = tk.Frame(
    root,
    bg="#111c2e",
    height=100
```

Ln: 52 Col: 0
28°C Partly sunny 08:31 10-08-2026

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```
Practical No. 4 Cyber Security.py - C:/Users/mvuc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

header.pack(fill="x")

header.pack_propagate(False)

title_label = tk.Label(
    header,
    text="🔐 RSA DIGITAL SIGNATURE",
    font=("Segoe UI", 22, "bold"),
    bg="#111c2e",
    fg="#ffffff"
)

title_label.pack(pady=(20, 0))

subtitle_label = tk.Label(
    header,
    text="Secure Message Authentication & Integrity Verification",
    font=("Segoe UI", 10),
    bg="#111c2e",
    fg="#7f8ea3"
)

subtitle_label.pack(pady=5)

Ln: 52 Col: 0
28°C Partly sunny
```

```
Practical No. 4 Cyber Security.py - C:/Users/mvuc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

content = tk.Frame(
    root,
    bg="#0b1220"
)

content.pack(
    fill="both",
    expand=True,
    padx=35,
    pady=25
)

message_title = tk.Label(
    content,
    text="MESSAGE",
    font=("Segoe UI", 10, "bold"),
    bg="#0b1220",
    fg="#00bcd4"
)

message_title.pack(anchor="w")

message_entry = tk.Entry(
    content,
    font=("Segoe UI", 12),
)

Ln: 52 Col: 0
28°C Partly sunny
```

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```
Practical No. 4 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

    bg="#162235",
    fg="ffffff",
    insertbackground="#00bcd4",
    relief="flat",
    bd=0
)

message_entry.pack(
    fill="x",
    ipady=12,
    pady=(8, 22)
)

signature_title = tk.Label(
    content,
    text="DIGITAL SIGNATURE",
    font=("Segoe UI", 10, "bold"),
    bg="#0b1220",
    fg="#00bcd4"
)

signature_title.pack(anchor="w")

signature_entry = tk.Text(
    content,
```

```
Practical No. 4 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

    height=9,
    font=("Consolas", 9),
    bg="#162235",
    fg="#d7e3f4",
    insertbackground="#00bcd4",
    relief="flat",
    bd=0,
    wrap="word"
)

signature_entry.pack(
    fill="x",
    pady=(8, 22)
)

button_frame = tk.Frame(
    content,
    bg="#0b1220"
)

button_frame.pack(pady=5)

sign_button = tk.Button(
    button_frame,
    text="🔑 Generate Signature",
```

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```
Practical No. 4 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

    command=sign_message,
    font=("Segoe UI", 10, "bold"),
    bg="#007c91",
    fg="white",
    activebackground="#0097a7",
    activeforeground="white",
    relief="flat",
    cursor="hand2",
    padx=18,
    pady=10
)

sign_button.grid(
    row=0,
    column=0,
    padx=7
)

verify_button = tk.Button(
    button_frame,
    text="✓ Verify Signature",
    command=verify_signature,
    font=("Segoe UI", 10, "bold"),
    bg="#087f5b",
    fg="white",
    activebackground="#099268",
    activeforeground="white",
    relief="flat",
    cursor="hand2",
    padx=18,
    pady=10
)

Ln: 52 Col: 0
28°C Partly sunny 10-08-2026
```

```
Practical No. 4 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

    cursor="hand2",
    padx=18,
    pady=10
)

verify_button.grid(
    row=0,
    column=1,
    padx=7
)

clear_button = tk.Button(
    button_frame,
    text="0 Clear",
    command=clear_fields,
    font=("Segoe UI", 10, "bold"),
    bg="#344054",
    fg="white",
    activebackground="#475467",
    activeforeground="white",
    relief="flat",
    cursor="hand2",
    padx=18,
    pady=10
)

clear_button.grid(
    row=0,
    column=2,
    padx=7
)

Ln: 52 Col: 0
28°C Partly sunny 10-08-2026
```

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```
Practical No. 4 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

    column=2,
    padx=7
)

status_frame = tk.Frame(
    content,
    bg="#111c2e",
    padx=20,
    pady=15
)

status_frame.pack(
    fill="x",
    pady=(25, 0)
)

status_label = tk.Label(
    status_frame,
    text="● READY",
    font=("Segoe UI", 11, "bold"),
    bg="#111c2e",
    fg="#00bcd4"
)

status_label.pack()
```

28°C Partly sunny

Ln: 52 Col: 0

08:34 10-08-2026

```
Practical No. 4 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

status_label.pack()

status_detail = tk.Label(
    status_frame,
    text="Enter a message and generate a digital signature.",
    font=("Segoe UI", 9),
    bg="#111c2e",
    fg="#9aa4b2"
)

status_detail.pack(pady=(5, 0))

footer = tk.Label(
    root,
    text="RSA-2048 • SHA-256 • PKCS#1 v1.5",
    font=("Segoe UI", 9),
    bg="#0b1220",
    fg="#667085"
)

footer.pack(
    pady=(0, 15)
)
```

28°C Partly sunny

Ln: 52 Col: 0

08:35 10-08-2026

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```
Practical No. 4 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 4 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help

text="Enter a message and generate a digital signature.",
font=("Segoe UI", 9),
bg="#111c2e",
fg="#9aa4b2"
)

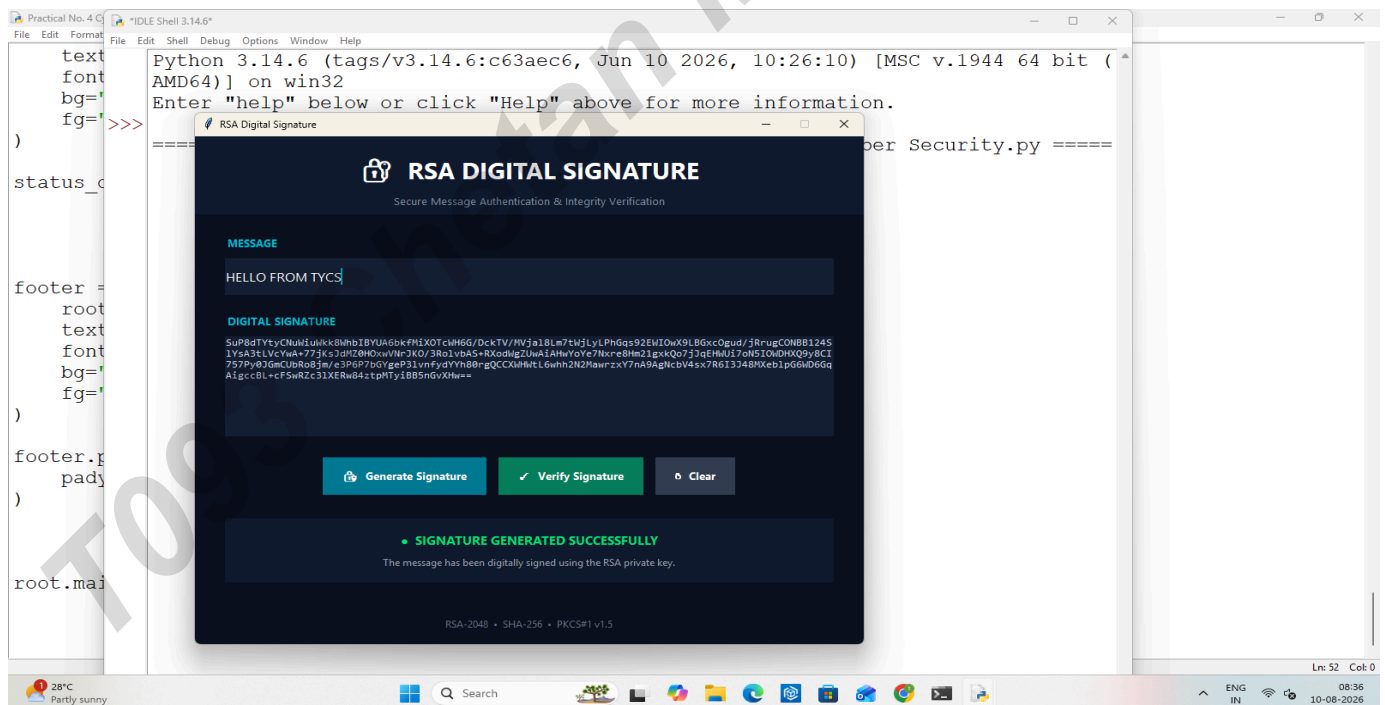
status_detail.pack(pady=(5, 0))

footer = tk.Label(
    root,
    text="RSA-2048 • SHA-256 • PKCS#1 v1.5",
    font=("Segoe UI", 9),
    bg="#0b1220",
    fg="#667085"
)

footer.pack(
    pady=(0, 15)
)

root.mainloop()
```

Output:-

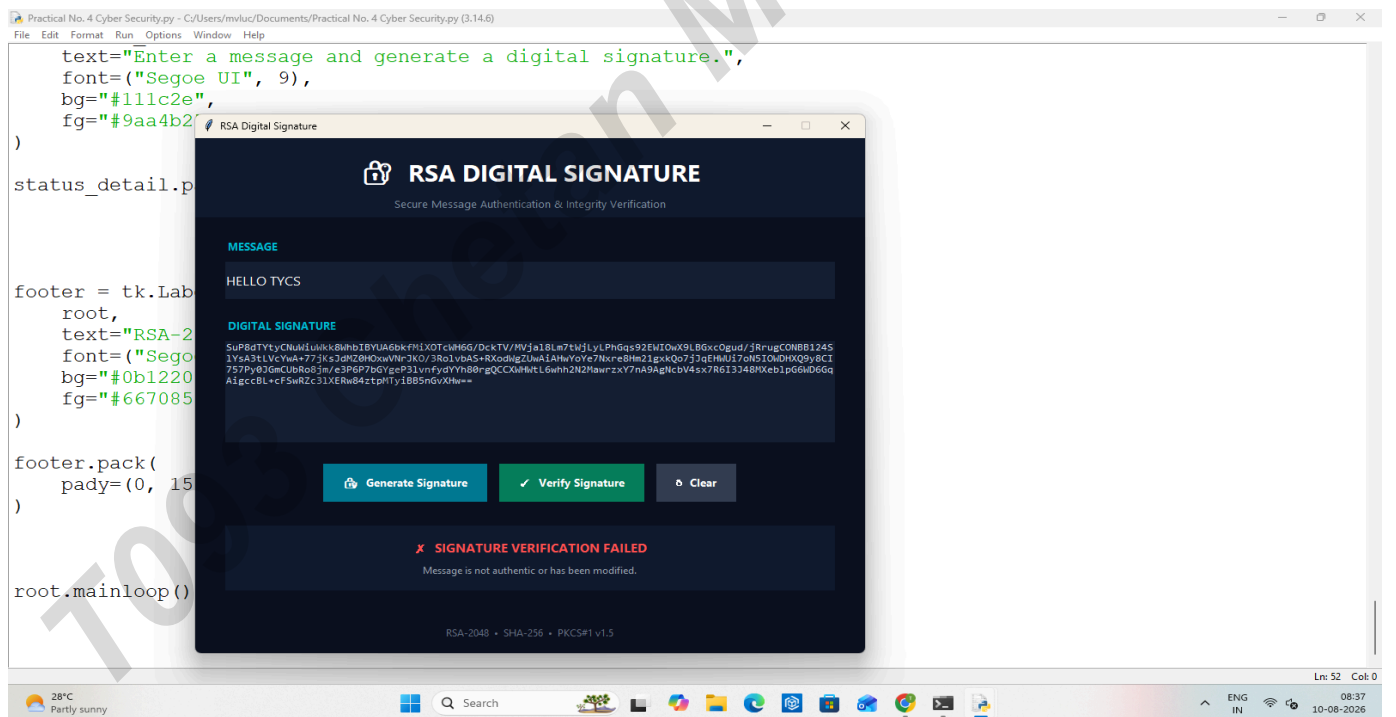
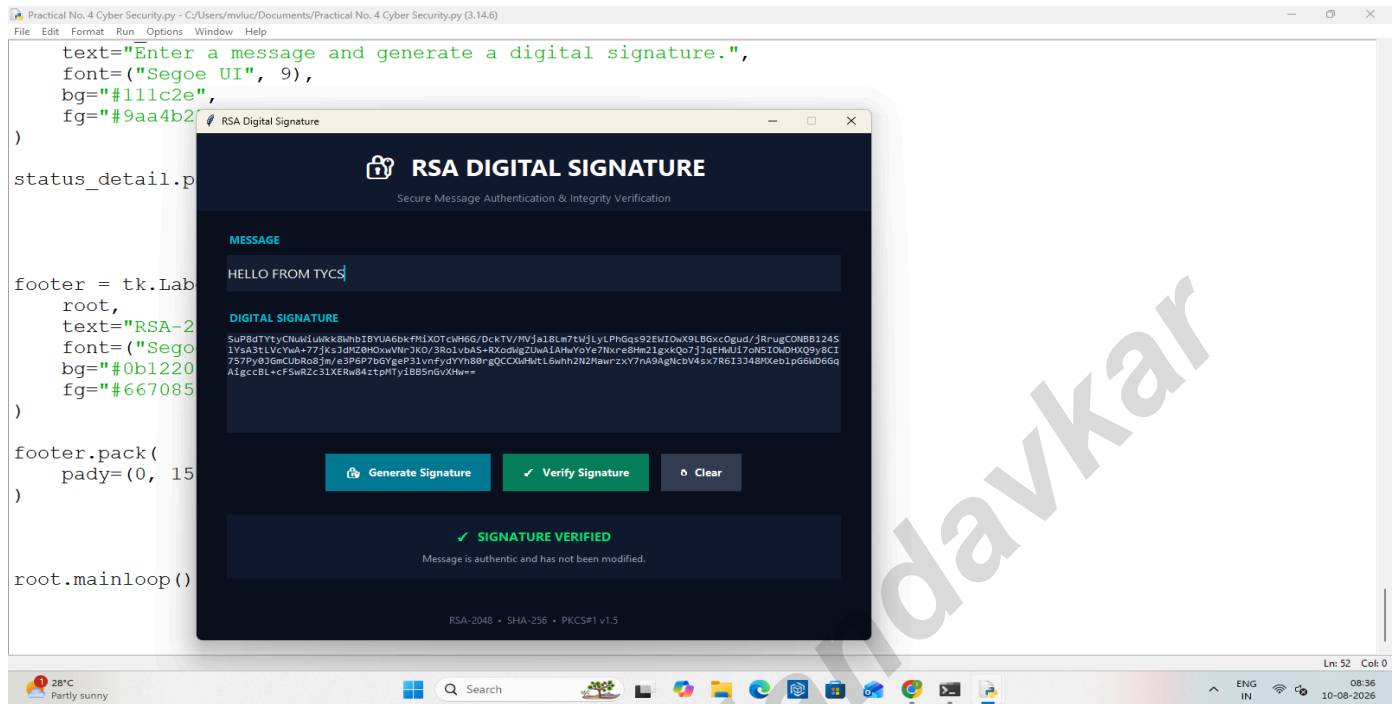


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